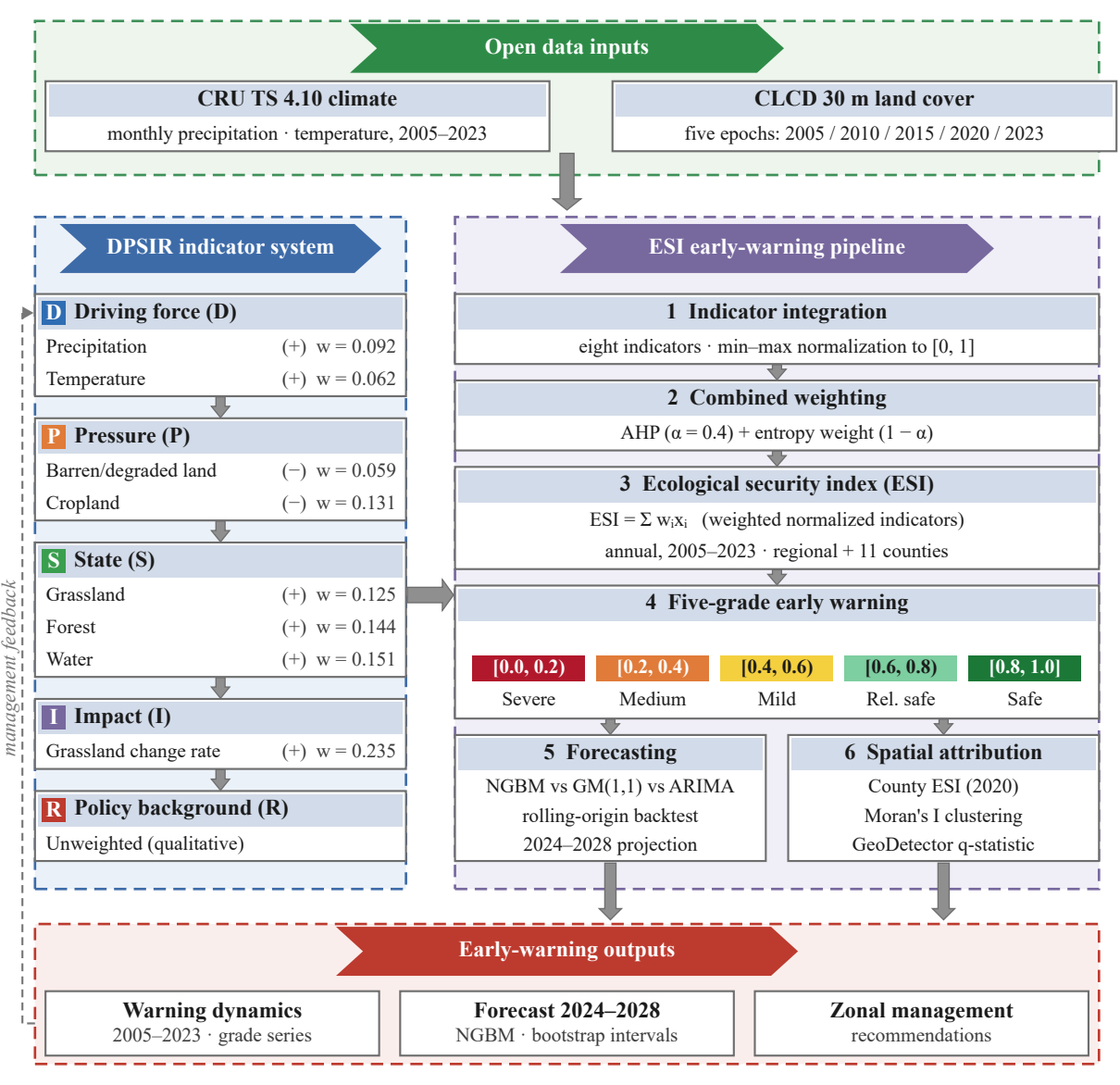

Open data inputs
CRU TS 4.10 climate
monthly precipitation · temperature, 2005–2023
CLCD 30 m land cover
five epochs: 2005 / 2010 / 2015 / 2020 / 2023
DPSIR indicator system
D Driving force (D)
Precipitation (+) w = 0.092
Temperature (+) w = 0.062
P Pressure (P)
Barren/degraded land (−) w = 0.059
Cropland (−) w = 0.131
S State (S)
Grassland (+) w = 0.125
Forest (+) w = 0.144
Water (+) w = 0.151
I Impact (I)
Grassland change rate (+) w = 0.235
R Policy background (R)
Unweighted (qualitative)
management feedback
ESI early-warning pipeline
1 Indicator integration
eight indicators · min–max normalization to [0, 1]
2 Combined weighting
AHP (α = 0.4) + entropy weight (1 − α)
3 Ecological security index (ESI)
ESI = Σ wᵢxᵢ (weighted normalized indicators)
annual, 2005–2023 · regional + 11 counties
4 Five-grade early warning
[0.0, 0.2) Severe
[0.2, 0.4) Medium
[0.4, 0.6) Mild
[0.6, 0.8) Rel. safe
[0.8, 1.0] Safe
5 Forecasting
NGBM vs GM(1,1) vs ARIMA
rolling-origin backtest
2024–2028 projection
6 Spatial attribution
County ESI (2020)
Moran's I clustering
GeoDetector q-statistic
Early-warning outputs
Warning dynamics
2005–2023 · grade series
Forecast 2024–2028
NGBM · bootstrap intervals
Zonal management
recommendations